Question

Ester compounds $\mathbf{A}$ and $\mathbf{B}$ undergo a condensation reaction in a solution of sodium ethoxide in ethanol. After the reaction, the mixture is treated with water, and one of the products is $\mathbf{C}$; the spectroscopic analysis results are shown in the table below:

Compound	$^1\mathrm{H-NMR}(\delta, \mathrm{ppm})$
$\mathbf{A}$	$8.03(1\mathrm{H}, \mathrm{s}), 4.22(2\mathrm{H}, \mathrm{q}), 1.29(3\mathrm{H}, \mathrm{t})$
$\mathbf{B}$	$4.12(2\mathrm{H}, \mathrm{q}), 2.04(3\mathrm{H}, \mathrm{s}), 1.26(3\mathrm{H}, \mathrm{t})$
$\mathbf{C}$	$8.82(3\mathrm{H}, \mathrm{s}), 4.42(6\mathrm{H}, \mathrm{q}), 1.41(9\mathrm{H}, \mathrm{t})$

Which of the following statements is correct:

A. All other options are incorrect

B. $\mathbf{B}$ Acid hydrolysis produces gas

C. $\mathbf{A, B, C}$ The hydrolysis products under acidic conditions are completely different.

D. $\mathbf{A, B, C}$ all belong to the C point group

E. $\mathbf{C}$ is capable of self-condensation.

F. $\mathbf{A}$, $\mathbf{B}$ The product of a single condensation reaction occurring between them is only one type.

Answer

Correct Answer: F

Detailed Explanation

Analyze the proton NMR spectrum:

Since they are all ester compounds, the alcohol before condensation has at least one methyl group; therefore, for $\mathbf{A, B, C}$, their $3\mathrm{H, t}$ peaks are all methyl peaks, and the high-field $2\mathrm{H, q}$ peak is methylene, suggesting that these five hydrogens constitute an ethyl group, so $\mathbf{A, B, C}$ all produce ethanol after hydrolysis. Option C is incorrect.

CHECKPOINT 1 PTS

$3\mathrm{H, t}$ peaks are all methyl peaks, and the high-field $2\mathrm{H, q}$ peak is methylene, constituting an ethyl group

CHECKPOINT 1 PTS

$\mathbf{A, B, C}$ all produce ethanol after hydrolysis

Observing the proton NMR spectrum of $\mathbf{C}$, it has a low-field peak, which can be speculated to be aromatic hydrogen. $\mathbf{C}$ has three ethyl groups, which can be speculated to be three ethyl ester groups and three aromatic hydrogens, so $\mathbf{C}$ contains a benzene ring; since the chemical shifts of the aromatic hydrogens are the same, their chemical environments are also the same, $\mathbf{C}$ is actually meta-tris(ethyl ester)benzene, with the structure of; O=C(C1=CC(C(OCC)=O)=CC(C(OCC)=O)=C1)OCC

CHECKPOINT 1 PTS

C is actually meta-tris(ethyl ester)benzene, with the structure of : O=C(C1=CC(C(OCC)=O)=CC(C(OCC)=O)=C1)OCC

A contains only one low-field hydrogen, which is unlikely to be aromatic hydrogen, considering it to be aldehyde hydrogen; therefore, **A** is ethyl formate, with the structure of O=C([H])OCC

CHECKPOINT 1 PTS

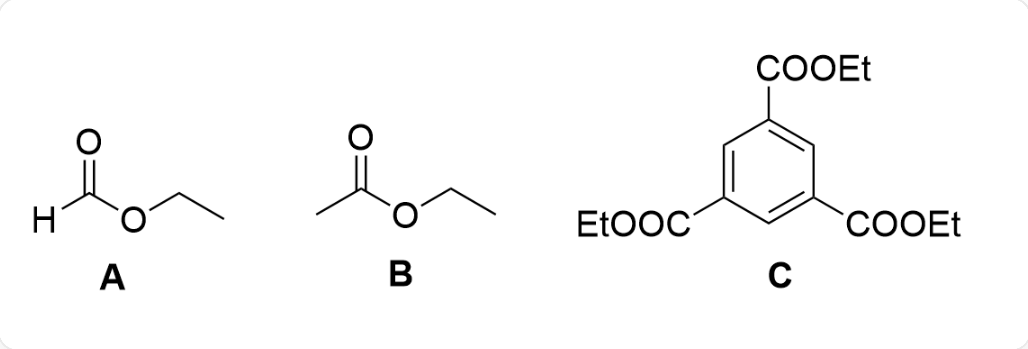
A is ethyl formate, with the structure of O=C([H])OCC

B does not contain low-field hydrogen but has an extra $3H, s$ peak, which is obviously the methyl group at the alpha position of the ester group. Therefore, **B** is ethyl acetate, with the structure of O=C(C)OCC

CHECKPOINT 1 PTS

B is ethyl acetate, with the structure of O=C(C)OCC

B produces acetic acid and ethanol after hydrolysis, option B is incorrect. Obviously, **C** belongs to the D point group, option D is incorrect. **C** has no α-H and cannot self-polymerize, option E is incorrect. **A** has no α-H, so there is only one ester condensation product, option F is correct.



Structure of $\mathbf{C}$: O=C(C1=CC(C(OCC)=O)=CC(C(OCC)=O)=C1)OCC; Structure of $\mathbf{A}$: O=C([H])OCC; Structure of $\mathbf{B}$: O=C(C)OCC